Rudrajit Choudhuri

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Updated November 5, 2026

Summary

Mixed-methods UX and HCI researcher with 6+ years of experience leading end-to-end research on Human-Centered AI, Software Engineering, Cognitive Science, UX Design, Future of Work, and CS Education. Two Microsoft internships (Microsoft Research, SAINTES; Microsoft Tech Futures, X3) delivering large-scale studies with 3,300+ participants, design frameworks, and product guidance adopted by partner teams. 33 publications at leading SE/HCI venues (ICSE, TOSEM, IST, ICSME, ICER, EMSE, and ACMQ) and Machine Learning and Computer Vision venues (CVIP, ICVGIP, Applied Soft Computing, and Neural Computing and Applications), with four distinguished paper awards. Research covered by media and featured in industry newsletters. Currently focused on (1) cognitive and socio-technical factors shaping human–AI collaboration, and (2) designing human-centered interfaces for AI tools that improve AI-native knowledge work.

Education

Oregon State University **Corvallis, OR**

Ph.D. in Computer Science (in progress), GPA 3.9/4.0. Focus: Human-Centered AI, Software Engineering

M.S. in Computer Science, GPA 4.0/4.0. Focus: Human-Centered AI, CS Education

Committee: A. Sarma (Advisor), M. Burnett, C. Hundhausen, I. Steinmacher, C. Sanchez

St. Thomas' College of Engineering & Technology **Kolkata, India**

B.Tech (Hons.) in Computer Science, GPA 9.81/10, Rank 1 in college

Advisor: A. Halder. Focus: Machine Learning & Deep Learning in Radiology, Biomedicine, and Robotics

Experience

Graduate Research Assistant **Sep 2022 – Present**

Oregon State University, Advisor: Anita Sarma *Corvallis, OR*

Except Summer 2024; Spring 2025 (fellowship); Summers 2025 and 2026 (Microsoft internships)

- Led user research on generative AI's effects on productivity, cognition, and self-efficacy; mapped AI failure modes and their causes to concrete design strategies for safer, more effective human–AI interaction (ICSE 2024).
- Built an empirically grounded model of the factors shaping developers' trust in and adoption of AI, through a large-scale mixed-methods study with developers at GitHub and Microsoft (N=238); published at ICSE 2025 and extended in ACM TOSEM.
- Modeled how routine AI reliance leads to cognitive debt among STEM students across five universities (N=300), producing interface and curriculum interventions that preserve reflection and critical thinking (IST 2026); covered by TV and news media.
- Led user research with CS students on where and why they struggle to use AI in learning, detailing impacts on learning, task quality, self-perception, and adoption decisions (CSEE&T 2025).
- Co-developed a framework modeling how interpersonal challenges affect the sense of welcomeness in open source, through a study of 706 Linux Foundation contributors, producing actionable practices for inclusive communities (ICSE 2025).

Applied Science Intern, Tech Futures X3 **Jun 2026 – Aug 2026**

Microsoft, Mentors: Alice Ferng, Cinoo Lee, Peter Jahn *Redmond, WA*

- Led a large-scale investigation into what makes a great co-worker in an AI-native workplace (22 interviews; mixed-methods survey of 1,534 knowledge workers). Built the BACI framework of co-worker qualities that apply to both humans and AI, identified 11 co-worker archetypes and how priorities vary across individuals, and a taxonomy of AI work etiquette for sharing AI-supported work.
- Transformed years of human-centered AI research into the Design Cookbook for UI/UX of AI, an interactive resource of 80 design patterns across 10 design dimensions mapped to 21 human-factors challenges, bridging research and product design.
- Led an empirical study with 448 developers on where and why they draw the line on AI autonomy; showed how task identity, accountability, and cognitive demands shape delegation decisions, and derived implications for agentic tooling and meaningful work design.
- Co-led an investigation into agent-to-agent coordination in large-scale multi-agent workplaces.

PhD Research Intern, SAINTES Group **Jun 2025 – Sep 2025**

Microsoft Research, Mentors: Christian Bird, Robert DeLine, Carmen Badea *Redmond, WA*

- Led a large-scale mixed-methods study with Microsoft developers (N=860) to identify where AI delivers real value in software engineering and how to design it responsibly so developers retain agency, craft, and meaningful work. Built a cognitive task-appraisal framework tied to AI openness and use, identified high-need/low-use targets for tooling investment, and specified 22 AI systems developers want built with the Responsible AI guardrails needed to ship them (ICSE-SEIP 2026; ICSME 2026 Distinguished Paper Award).
- Conducted eight large-scale experiments with 2,000 developers testing contextual transparency patterns and cognitive-forcing interventions; showed that targeted friction improved verification and reduced over-reliance while preserving UX, and produced causal design guidelines for appropriate reliance.
- Co-designed a matched-pair study of individual and team AI usage patterns; identified the Productivity Pressure Paradox (rising expectations without enablement erode benefits) and delivered organizational and product levers to unlock durable gains (ICSE 2026 Distinguished Paper Award).
- Translated findings into product practices with AI for Healthcare, AI Ethics, and Copilot Bing Search, delivering actionable guidance and evaluation rubrics adopted by partner teams.

MOSIP Global Impact Research Fellow (Gates Foundation-funded fellowship) **Mar 2025 – Jun 2025**
Modular Open Source Identity Platform (MOSIP), IIIT Bangalore *Corvallis, OR*

- Designed an end-to-end multi-agent workflow to automatically debug usability issues in national ID systems; operationalized inclusivity decision rules into AI-driven checks, achieving an 89% F1 score when validated on academic courseware and industry products.
- Built an extensible framework adaptable to diverse personas and inclusivity dimensions (cognitive, socio-economic, gender), providing actionable recommendations to improve equity in large-scale national identity platforms.

Research Scholar, AI4ICPS Lab **May 2022 – Jul 2022**
Indian Institute of Technology (IIT) Kharagpur, Advisor: Manoj Sharma *Kharagpur, India*

- Led research on trustworthy cyber-physical systems; developed structure-based model defenses against adversarial attacks on autonomous driving robots (CVIP 2022).

Research Student (REU) **Nov 2021 – Apr 2022**
Oregon State University, Advisor: Anita Sarma *Remote*

- Co-developed the Automated Inclusivity Detector (AID) tool to detect cognitive-bias bugs that impact diverse students in online CS courseware, and co-designed the user study evaluating it; fixes significantly improved students' control over their learning experience (ICER 2022, ICER 2024).

Associate Software Developer Intern **Sep 2021 – Dec 2021**
Nomura Research Institute Financial Technologies India Pvt. Ltd. *Kolkata, India*

- Co-developed an automated reconciliation solution enabling extensible, configurable processes for large multi-format databases, improving functionality and efficiency across business operations.

Research Student **Mar 2019 – Jul 2022**
St. Thomas' College of Engineering & Technology, Advisors: Amiya Halder, Amit Paul *Kolkata, India*

- Developed image processing, soft computing, statistical machine learning, and deep learning algorithms for biomedical and radiological diagnostics, satellite imaging, and robotics; two distinguished paper awards and a Neural Computing and Applications journal article.

Skills

UX Research Methods: Mixed-Methods Research, Field and User Studies, Survey Design and Validation, Semi-Structured Interviews, Contextual Inquiry, Usability Testing, A/B Testing, Experimental and Quasi-Experimental Design, Factorial and Vignette Studies, Longitudinal and Case Studies, Heuristic Evaluation, Cognitive Walkthrough, Inclusive Design, Personas, Participatory Design, Product and Market Research, Grounded Theory, Thematic Analysis, Qualitative Coding

Quantitative and Causal Analysis: Hypothesis Testing, Regression Analysis, Multilevel and Mixed-Effects Models, PLS-SEM, CB-SEM, Exploratory and Confirmatory Factor Analysis, Psychometric Scale Development and Validation, Mediation and Moderation Analysis, Bayesian Statistics, Causal Inference (Randomized Experiments, Quasi-Experimental Methods), Statistical Modeling, Data Mining of Software Repositories and Telemetry

Machine Learning and Generative AI: Supervised and Unsupervised Learning, Feature Engineering, Deep Learning (ANN, CNN, RNN, Transformers, Attention, Adversarial Networks), NLP, Computer Vision, Image Segmentation, LLM Fine-Tuning, Prompt Engineering, LLM Evaluation and Harness Engineering, LLM-as-a-Judge, RAG, Neural Retrieval, Agentic and Multi-Agent Workflows, Agent Orchestration and Coordination, Tool Use/Function Calling, LangGraph, MCP, A2A

Human-Centered AI Design: AI UX Design Patterns, Human–AI Interaction Guidelines, Appropriate Trust and Reliance, Transparency and Explainability, Responsible AI Evaluation, Cognitive Engagement, Work Design and Task Appraisal, Design Systems and Pattern Libraries

Programming, Frameworks, and Tools: Python, R, C/C++, Java, JavaScript, TypeScript, SQL, MATLAB, LaTeX, HTML/CSS, Scikit-Learn, TensorFlow, PyTorch, Keras, OpenCV, Flask, React, Node.js, Git, AWS, Qualtrics, SurveyMonkey, Atlas.ti, Figma, RStudio, JASP, SmartPLS

Selected Publications

35 total publications: 21 conference papers, 6 journal articles, 1 magazine article, 1 book chapter, 6 preprints/under review. Four distinguished paper awards (🏆). Full list on Google Scholar; [pdf] and [data] are hyperlinks.

Preprints and Under Review

The Vibe Paradox: What EEG Reveals About Your Brain When Vibe Coding [pdf | data] **Preprint, 2026**
O Schutz*, **R Choudhuri***, Y Qiao, FA Moussaoui, J Russo, Z Steine-Hanson, A Sarma *Co-first authors; EEG, n=11

"To Boldly Go Where No One Has Gone Before": What Has AI Let Developers Unlock? [pdf | data] **Preprint, 2026**
R Choudhuri, JL Butler, C Bird, C Badea, B Houck, A Sarma Microsoft, N=331 developers

What Makes a Great Co-Worker in an AI-Native Workplace? [pdf | data] **Under Review, 2026**
R Choudhuri, M Meijer, SYT Lee, C Lee, C Mannion, P Jahn, A Sarma, C Bird, A Ferng Microsoft, N=1,556 knowledge workers

You Shall Not Pass! Where and Why Developers Draw the Line on AI Autonomy [pdf | data] **Under Review, 2026**
R Choudhuri, C Bird, C Badea, M Gerosa, A Sarma Microsoft, N=448 developers

Journal and Magazine Articles

What Needs Attention? Prioritizing Drivers of Developers' Trust and Adoption of Generative AI [pdf | data] — **ACM TOSEM 2026**
R Choudhuri, B Trinkenreich, R Pandita, E Kalliamvakou, I Steinmacher, M Gerosa, C Sanchez, A Sarma — Impact Factor: 6.9

Thinking Less, Trusting More: GenAI's Impacts on Students' Cognitive Habits [pdf | data] — **IST 2026**
R Choudhuri, C Sanchez, M Burnett, A Sarma — Impact Factor: 4.6

The AI-Native Developer: Redefining Work, Identity, and the Future of Craft [pdf] — **ACM Queue 2026**
R Choudhuri, E Kalliamvakou, B Houck, T Zimmermann — Vol. 24, No. 2

Adaptive Rough Fuzzy Conditional Spatial Clustering Algorithms for Noisy Brain Tissue Segmentation [pdf] — **Applied Soft Computing 2026**
A Halder, **R Choudhuri**, A Sarkar — Impact Factor: 7.8

Measuring SES-Related Traits Relating to Technology Usage: Two Validated Surveys [pdf] — **EMSE 2025**
C Chikezie, P Chanpaisaeng, P Agarwal, S Afroz, B Madhwani, **R Choudhuri**, A Anderson, P Velhal, P Morreale, C Bogart, A Sarma, M Burnett — Impact Factor: 4.3

Brain MRI Tumour Classification Using Quantum Classical Convolutional Neural Network Architecture [pdf] — **NCAA 2023**
R Choudhuri, A Halder — Impact Factor: 4.5

Conference Papers

🏆 **To Copilot and Beyond: 22 AI Systems Developers Want Built [pdf | data]** — **ICSME-Industry 2026**
R Choudhuri, C Bird, C Badea, A Sarma. **[IEEE Distinguished Paper Award]** — Acceptance: 37%

🏆 **"Maybe We Need Some More Examples": Individual and Team Drivers of Developer GenAI Tool Use [pdf | data]** — **ICSE 2026**
C Miller, **R Choudhuri**, M Ulloa, S Haniyur, R DeLine, MA Storey, E Murphy-Hill, C Bird, J Butler — Acceptance: 21.8%
[ACM SIGSOFT Distinguished Paper Award]

AI Where It Matters: Where, Why, and How Developers Want AI Support in Daily Work [pdf | data | dashboard] — **ICSE-SEIP 2026**
R Choudhuri, C Badea, C Bird, J Butler, R DeLine, B Houck — Acceptance: 37%

What Guides Our Choices? Modeling Developers' Trust and Behavioral Intentions Towards GenAI [pdf | data] — **ICSE 2025**
R Choudhuri, B Trinkenreich, R Pandita, E Kalliamvakou, I Steinmacher, M Gerosa, C Sanchez, A Sarma — Acceptance: 21.2%

Investigating the Impact of Interpersonal Challenges on Feeling Welcome in OSS [pdf] — **ICSE 2025**
B Trinkenreich, Z Feng, **R Choudhuri**, M Gerosa, A Sarma, I Steinmacher — Acceptance: 21.2%

Insights from the Frontline: GenAI Utilization Among Software Engineering Students [pdf | data] — **CSEE&T 2025**
R Choudhuri, A Ramakrishnan, A Chatterjee, B Trinkenreich, I Steinmacher, M Gerosa, A Sarma — Acceptance: 31%

How Far Are We? The Triumphs and Trials of Generative AI in Learning Software Engineering [pdf | data] — **ICSE 2024**
R Choudhuri, D Liu, I Steinmacher, M Gerosa, A Sarma — Acceptance: 21.2%

Debugging for Inclusivity in Online CS Courseware: Does It Work? [pdf | data] — **ICER 2024**
A Chatterjee, **R Choudhuri**, M Sarkar, S Chattopadhyay, D Liu, S Hedaoo, M Burnett, A Sarma — Acceptance: 20.1%

Structure-Based Learning for Defense Against Adversarial Attacks in Autonomous Driving Agents [pdf] — **CVIP 2022**
MK Sharma, **R Choudhuri**, M Dixit, M Sarkar, B Dittakavi — Acceptance: 33%

🏆 **Adaptive Rough-Fuzzy Kernelized Clustering Algorithm for Noisy Brain MRI Tissue Segmentation [pdf]** — **CVIP 2021**
R Choudhuri, A Halder. **[IAPR Distinguished Paper Award]** — Acceptance: 26.1%

🏆 **High-Density Salt and Pepper Noise Removal Algorithm Using Statistical Approach [pdf]** — **ICACA 2021**
A Halder, **R Choudhuri**. **[Best Paper Award]** — Acceptance: 29.7%

Book Chapter

Automated Brain Tumor Analysis Using Deep Learning Based Framework [pdf] — **CRC Press 2023**
A Halder, A Sarkar, **R Choudhuri**. In: Medical Data Analysis and Processing Using Explainable AI

Awards, Grants, and Honors

- 🏆 ACM SIGSOFT Distinguished Paper Award, ICSE 2026
- 🏆 IEEE Distinguished Paper Award, ICSME 2026 (Industry Track)
- 🏆 IAPR Distinguished Paper Award, CVIP 2021
- 🏆 Best Paper Award, ICACA 2021
- MOSIP Global Impact Inclusive Research Fellowship (Gates Foundation-funded), 2025
- NSF Travel Award, ICSE 2024 and ICSE 2025
- OpenAI Researcher Access Program Grant, PI, 2025
- 🏅 Gold Medalist, NPTEL (IIT Ropar), 2020; Rank 1 in college, B.Tech Computer Science, 2022

Invited Talks and Presentations

What Makes a Great Co-Worker in an AI-Native Workplace? **Aug 2026**
Internship Talk, Tech Futures Group, Microsoft

AI Where It Matters: Where, Why, and How Developers Want AI Support in Daily Work **Sep 2025**
Internship Talk, SAINTES Group, Microsoft Research

What Needs Attention? Designing for Appropriate Trust in AI **Jul 2025**
Invited Talk, Appropriate Reliance Group, Microsoft Research

What Guides Our Choices? Modeling Developers' Trust and Adoption Towards GenAI **May 2025**
International Conference on Software Engineering (ICSE) 2025

Insights from the Frontline: GenAI Utilization Among Software Engineering Students **Apr 2025**
IEEE Conference on Software Engineering Education and Training (CSEE&T) 2025

Cognitive Factors Affecting Trust and Adoption Towards AI **Nov 2024**
Invited Talk, Colorado State University

How Far Are We? The Triumphs and Trials of Generative AI in SE **Apr 2024**
International Conference on Software Engineering (ICSE) 2024

Selected Media and Industry Coverage

Media Coverage

Study shows AI may inhibit cognitive abilities (15-minute interview) Oct 2026
KPTV FOX 12 Oregon (Link)

Tech-savvy STEM students prone to letting AI do their thinking for them, study shows Oct 2026
Oregon State University Newsroom (Link)

AI might be reshaping how college students learn, at the expense of critical thinking May 2026
KGW TV News (Link)

OSU study raises concerns about AI's impact on student thinking skills May 2026
KATU ABC2 TV News (Link)

OSU study warns of declining cognitive ability as AI use grows in STEM fields May 2026
KOIN 6 News (Link); Yahoo News (Link)

The staggering 'cognitive debt' of generative AI in education May 2026
Jefferson Public Radio, The Jefferson Exchange (Link)

America's STEM students are falling into an 'AI dependence spiral' May 2026
SAN News (Link)

Tech-savvy STEM students prone to letting AI do their thinking for them May 2026
The News Guard (Link)

AI literacy to prevent scams in the AI era May 2026
KGW TV News (Link)

Industry Coverage

AI made developers faster. Is their judgment keeping up? Oct 2026
Engineering Enablement (DX), Brian Houck; features "To Boldly Go Where No One Has Gone Before" (Link)

What makes a great co-worker in an AI-native workplace? Sep 2026
Research-Driven Engineering Leadership (RDEL #162), Lizzie Matusov; features "What Makes a Great Co-Worker in an AI-Native Workplace?" (Link)

Where do developers draw the line on AI autonomy? Aug 2026
Research-Driven Engineering Leadership (RDEL #158); features "You Shall Not Pass" (Link)

Five studies that are changing how I think about AI in software engineering Jul 2026
Engineering Enablement (DX), Brian Houck; features "AI Where It Matters" and "To Copilot and Beyond" (Link)

The AI-native developer May 2026
Engineering Enablement (DX), Brian Houck; features "The AI-Native Developer", ACM Queue (Link)

How does deepening AI fluency change what it means to be a software developer? May 2026
Research-Driven Engineering Leadership (RDEL #143); features "The AI-Native Developer" (Link)

Where do developers actually want AI to support their work? Oct 2025
Research-Driven Engineering Leadership (RDEL #114); features "AI Where It Matters" (Link)

What distinguishes developers who successfully adopt AI tools from those who don't? Sep 2025
Research-Driven Engineering Leadership (RDEL #108); features "Maybe We Need Some More Examples" (Link)

What influences developers' trust in adopting AI-assisted coding tools? Mar 2025
Research-Driven Engineering Leadership (RDEL #84); features "What Guides Our Choices?" (Link)

Professional Service and Mentoring

Mentoring, graduate students: Ambareesh Ramakrishnan, Sadia Afroz, Olivia Schutz, Fatima A. Moussaoui, Yunhan Qiao, Chifang Chou

Mentoring, undergraduate students: Dylan Liu, Mrinmoy Sarkar, Pierce Fleming, Arinjay Bhowmick, Mayank Dixit

Reviewer: IEEE Transactions on Software Engineering (TSE); Information and Software Technology (IST); Journal of Systems and Software (JSS); ACM CHI 2024; ACM Indian Conference on Computer Vision, Graphics and Image Processing (ICVGIP) 2022, 2023, 2024; International Conference on Computer Vision and Image Processing (CVIP) 2022, 2023

Sub-reviewer: ICPC 2025; ICSE 2024; MSR 2024; ESEC/FSE 2023; TSE; EMSE